

Supplementary File 4

User Operation Guide for the AIF (Artificial Intelligence for Food) Software

1. Overview

The AIF software is a standalone graphical user interface (GUI) developed to predict the shelf-life of marinated meat products and retrain machine learning models using new datasets. The interface consists of three primary functional modules: Home, Predict, and Retrain.

2. System Requirements and Startup

2.1 Environment: Windows OS (64-bit).

2.2 Startup: Double-click AIF.exe in the “dist” fold to launch the application. No installation of Python or external libraries is required.

3. Module 1: Home Interface

Upon launching, the Home interface serves as the central navigation hub (as shown in Figure S4.1).

(1) Navigation Bar:

Users can switch between modules by clicking the color-coded buttons: Home (Blue), Predict (Green), and Retrain (Orange).

(2) Module Preview:

The main area displays schematic diagrams illustrating the workflows for both shelf-life prediction and model retraining.

(3) Quick Start:

Click the “Start Predict” or “Start Retrain” buttons at the bottom to jump directly to the respective functions.

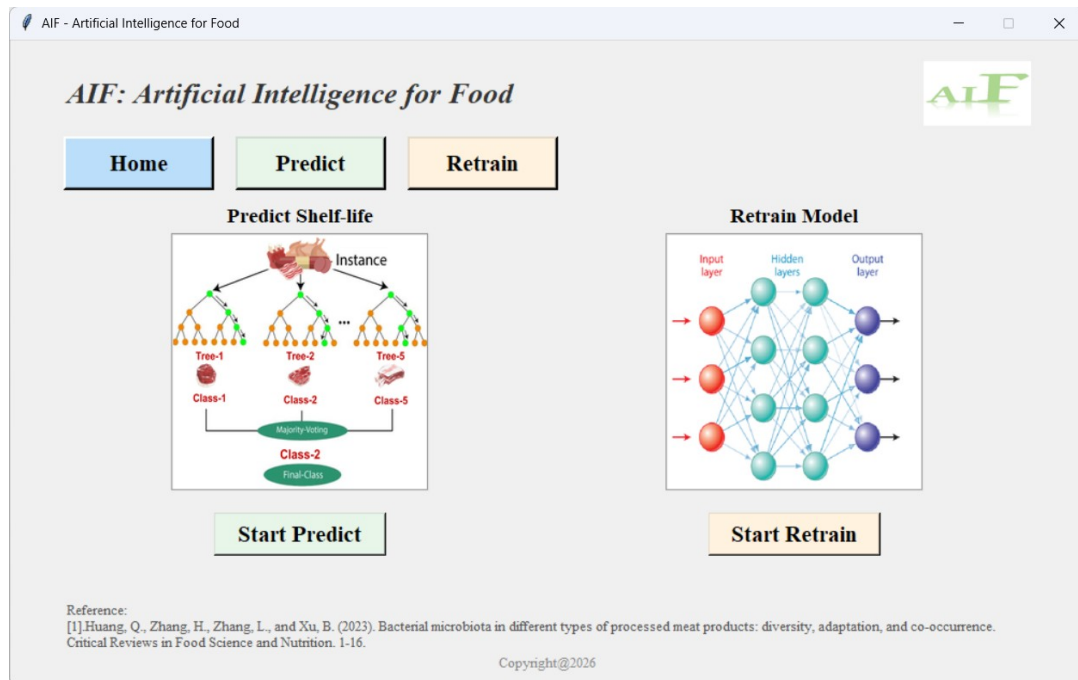


Figure S4.1. Home Interface

4. Module 2: Prediction Module

This module allows users to estimate shelf-life based on processing parameters.

(1) **Parameter Input** (as shown in Figure S4.2 (a)):

On the left panel (“Input Model Parameters”), select values from the dropdown menus for the four key features:

Packing Method: (e.g., Plain, Vacuum, Modified Atmosphere).

Preservatives: (e.g., None, Nisin, Chitosan).

Storage Temperature: (Low-temperature, Ambient, >25°C).

Secondary Sterilization: (e.g., None, Microwave, High-temp).

(2) **Model Selection** (as shown in Figure S4.2 (a)):

The system uses the pre-trained best_random_forest_model.pkl by default. Users can click “Load Custom Model” to import a “different.pkl” model file.

(3) **Execution** (as shown in Figure S4.2 (a)):

Click the green “PREDICT” button to generate results.

(4) Output Display (as shown in Figure S4.2 (b)):

The right panel displays:

(a) Shelf-life Class:

A visual icon indicating the predicted shelf-life range (e.g., “11 ~ 20 day”).

(b) Prediction Analysis:

A structured, bulleted list providing actionable recommendations for extending shelf-life based on the input parameters (e.g., suggesting specific sterilization methods or packaging improvements).

(a)

The screenshot shows a web application titled "AIF: Artificial Intelligence for Food". It features a navigation bar with three buttons: "Home" (light blue), "Predict" (green), and "Retrain" (orange). Below the navigation bar, there is a section titled "Input Model Parameters" with a "Load Custom Model" button. This section contains four dropdown menus: "Packing Method" (set to "Plain Packaging"), "Preservatives" (set to "None"), "Storage Temperature" (set to "Low-temperature (0~10°C)"), and "Secondary Sterilization" (set to "None"). A large green "PREDICT" button is positioned below these inputs. To the right of the input section is a large yellow box with the text "Ready to predict... Please select parameters on the left." At the bottom of the page, there is a reference list and a copyright notice.

AIF: Artificial Intelligence for Food

Home Predict Retrain

Input Model Parameters

Load Custom Model

Packing Method: Plain Packaging

Preservatives: None

Storage Temperature: Low-temperature (0~10°C)

Secondary Sterilization: None

PREDICT

Ready to predict...
Please select parameters on the left.

Reference:
[1] Huang, Q., Zhang, H., Zhang, L., and Xu, B. (2023). Bacterial microbiota in different types of processed meat products: diversity, adaptation, and co-occurrence. Critical Reviews in Food Science and Nutrition. 1-16.

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(b)

The screenshot shows a web application window titled "AIF - Artificial Intelligence for Food". The interface has a navigation bar with three buttons: "Home" (light blue), "Predict" (green), and "Retrain" (orange). Below the navigation bar, there are two main sections. The left section, titled "Input Model Parameters", contains a "Load Custom Model" button and four dropdown menus: "Packing Method" (set to "Plain Packaging"), "Preservatives" (set to "None"), "Storage Temperature" (set to "Low-temperature (0~10°C)"), and "Secondary Sterilization" (set to "None"). A large green "PREDICT" button is at the bottom of this section. The right section, titled "Prediction Analysis", features a green pentagon icon with the number "11" and a grey box displaying "11 ~ 20 day". Below this, the text "Prediction Analysis:" is followed by three bullet points: "• Vacuum or Modified Atmosphere Packaging may improve shelf life.", "• Sterilization methods (low/high temp, microwave) are recommended.", and "• Consider adding preservatives to extend shelf life." At the bottom of the window, there is a reference list and a copyright notice: "Reference: [1].Huang, Q., Zhang, H., Zhang, L., and Xu, B. (2023). Bacterial microbiota in different types of processed meat products: diversity, adaptation, and co-occurrence. Critical Reviews in Food Science and Nutrition. 1-16. Copyright@2026".

Figure S4.2. Prediction Module

5. Module 3: Retraining Module

This module enables the integration of new data to update the predictive model, featuring a robust data validation mechanism.

(1) Data Input (as shown in Figure S4.3 (a)):

Users have two methods to input new data:

- (c) Manual Input: Directly type or paste data into the text box.
- (d) File Upload: Click the orange "Upload CSV" button to select a "local .csv" file.

(2) Data Format Requirements:

- (a) The input must follow a standard Comma-Separated Values (CSV) format with the following header: temperature,package,preservative,sterilization,target
- (b) Values must be integers corresponding to the taxonomy defined in Table S1.

(3) Data Validation (as shown in Figure S4.3 (b)):

Upon input or upload, the system automatically checks for:

- (a) Header Consistency: Ensuring all five required columns are present.
- (b) Value Validity: Verifying that integers fall within allowed ranges (e.g., Package: 1-3).
- (c) Error Handling: If invalid data is detected, a “Data Validation Failed” dialog appears (as shown in the screenshots), specifying the error and providing a correct format example in large, bold text.

(4) Model Update (as shown in Figure S4.3 (c)):

If passed the “**Data Validation**” step, Click the blue “**SUBMIT**” button to fuse the new data with the existing dataset and retrain the Random Forest model.

(5) Performance Verification:

(a) Success Message (as shown in Figure S4.3 (d)):

A pop-up confirms successful training and displays the validation metrics (F1-score and Accuracy) of the new model.

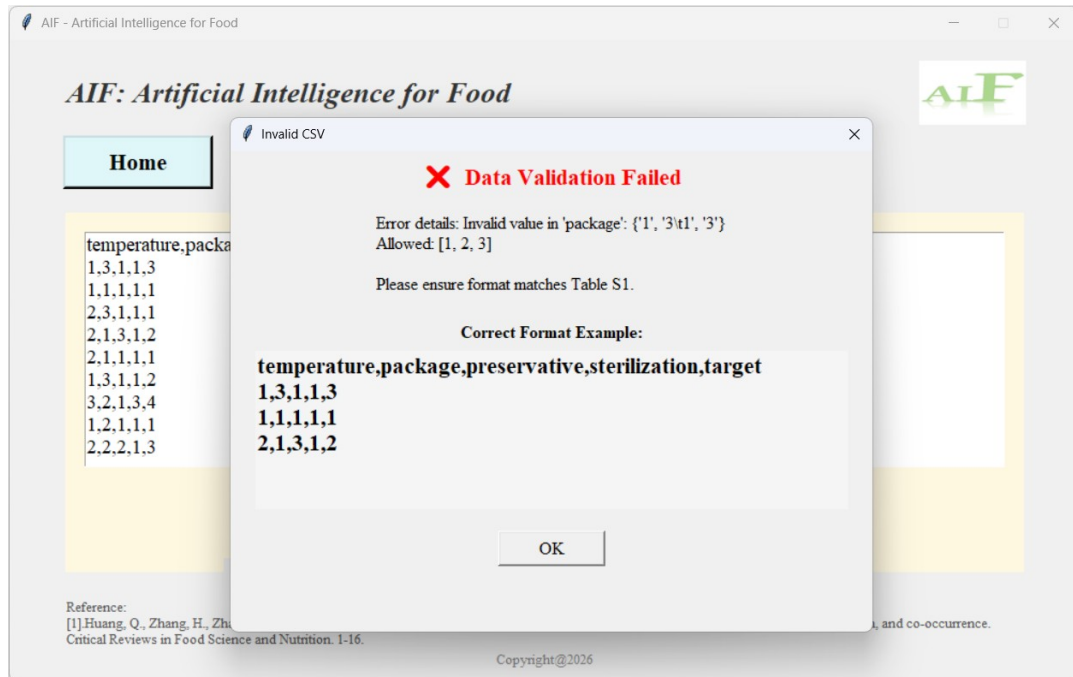
(b) Visual Validation (as shown in Figure S4.3 (e)):

A hyperlink “[Click here to view ROC Curve]” appears at the bottom. Clicking it opens the Receiver Operating Characteristic (ROC) curve image, allowing users to visually assess the model’s classification performance.

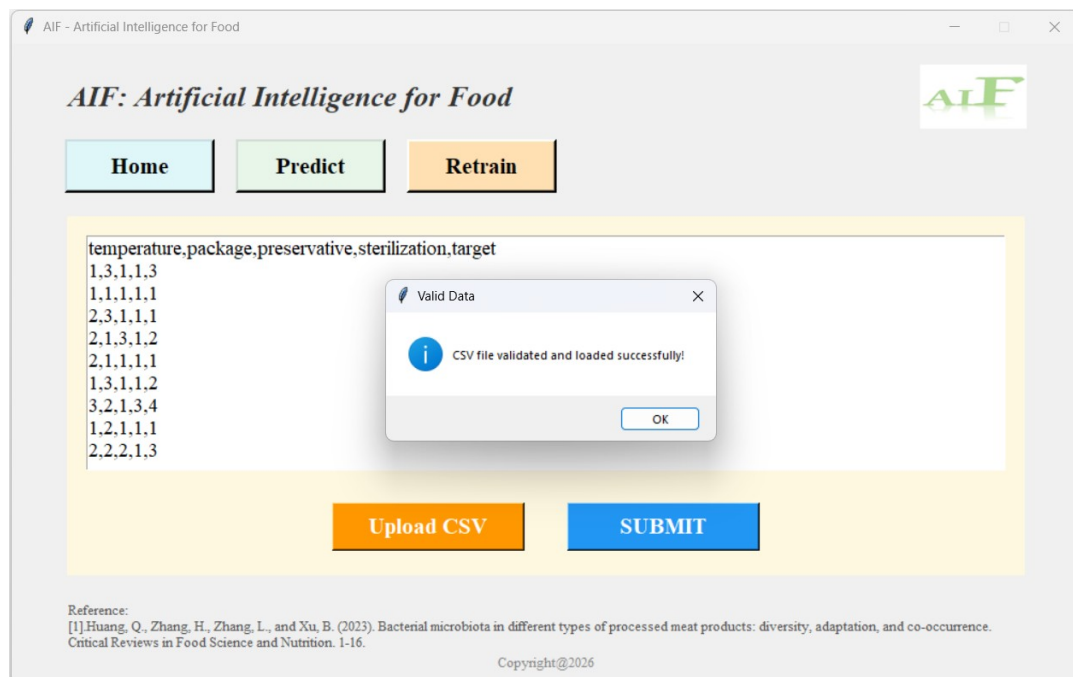
(a)



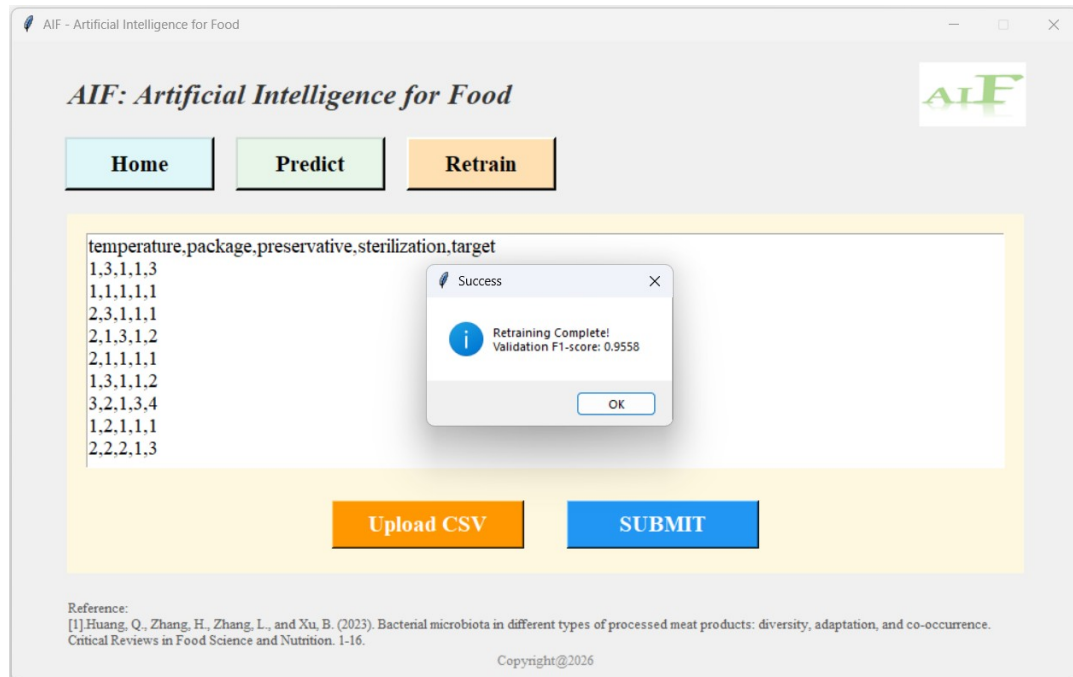
(b)



(c)



(d)



(e)

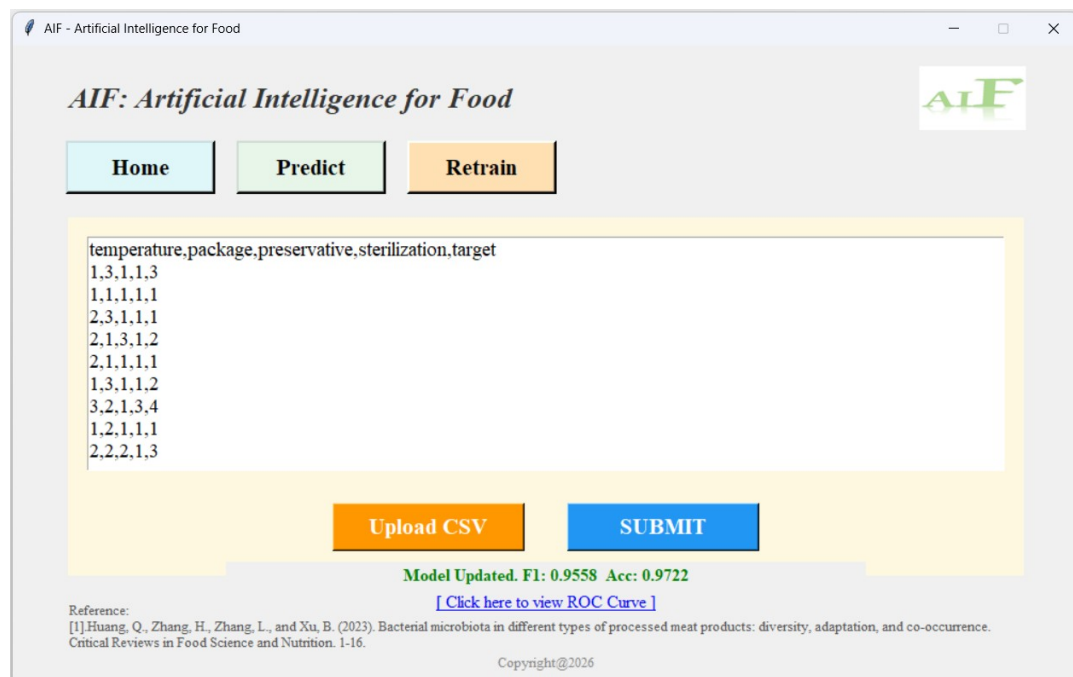


Figure S4.3. Retraining Module